Chapter 2



PLC Hardware Components

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Objectives

- MLO #2-17: Understand and apply the addressing convention for A-B and Click PLCs.
- MLO #2-18: Convert RLD int PLC ladder diagrams.
- MLO #2-19: Name the rules to draw a PLC ladder diagram and apply them in ladder programming.

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Section 2.1

I/O Addressing

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Addressing Inputs and Outputs

- All inputs and outputs must be wired to the terminals of the corresponding I/O modules.
- Each of these inputs have an address that depends on the module and the port they are connected to.
- In programming, we use this address in the ladder diagram.

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Addressing I/Os using A-B SLC 500

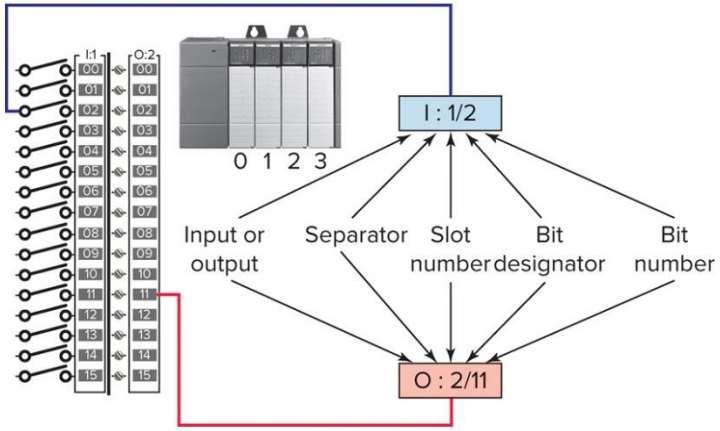
- In modular PLCs, slot 0 is reserved for processor.
- Input and output modules can be in any other slot.
- The address of an I/O point depends on the module and the port.
- The module refers to the slot and the port refers to the terminal number (bit if we see a port as a word)

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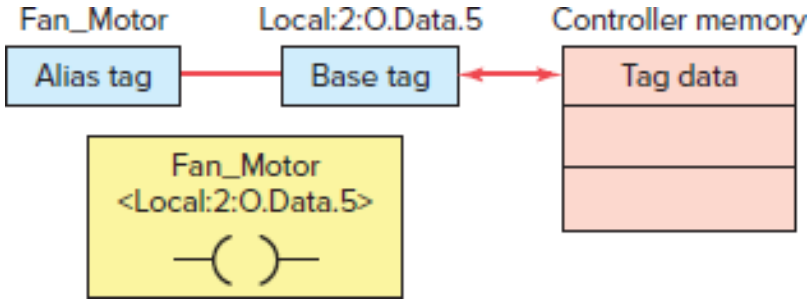
I/O Address: The A-B Case

An *address* is a label or number that indicates where a certain piece of information is located in memory.

Rack/slot-based addressing schemes are used with SLC 500 controllers.



Tag-based addressing is used with Allen-Bradley ControlLogix controllers. (Not used in this class)

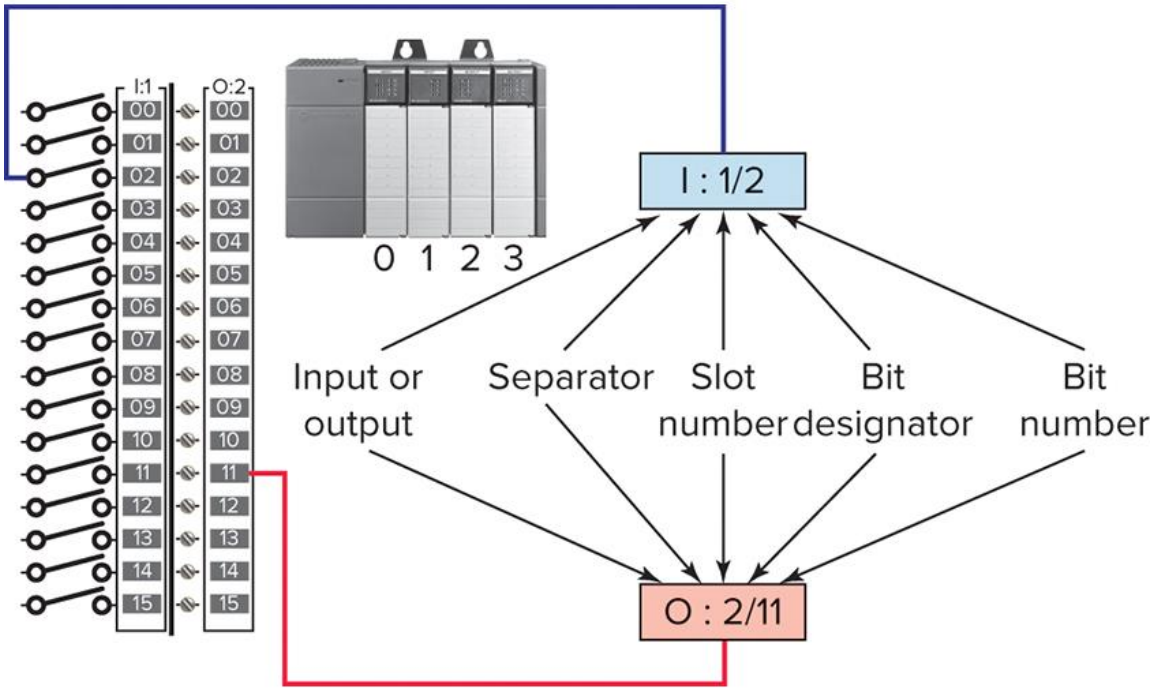


Access the text alternative for slide images.

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(Top Image) Source: Image Courtesy of Rockwell Automation, Inc.

Logix 500 *rack/slot-based* addressing format.



O:4/15 - Output module in slot 4, terminal 15

I:3/8 - Input module in slot 3, terminal 8

O:6/0 - Output module, slot 6

I:5/0 - Input module, slot 5

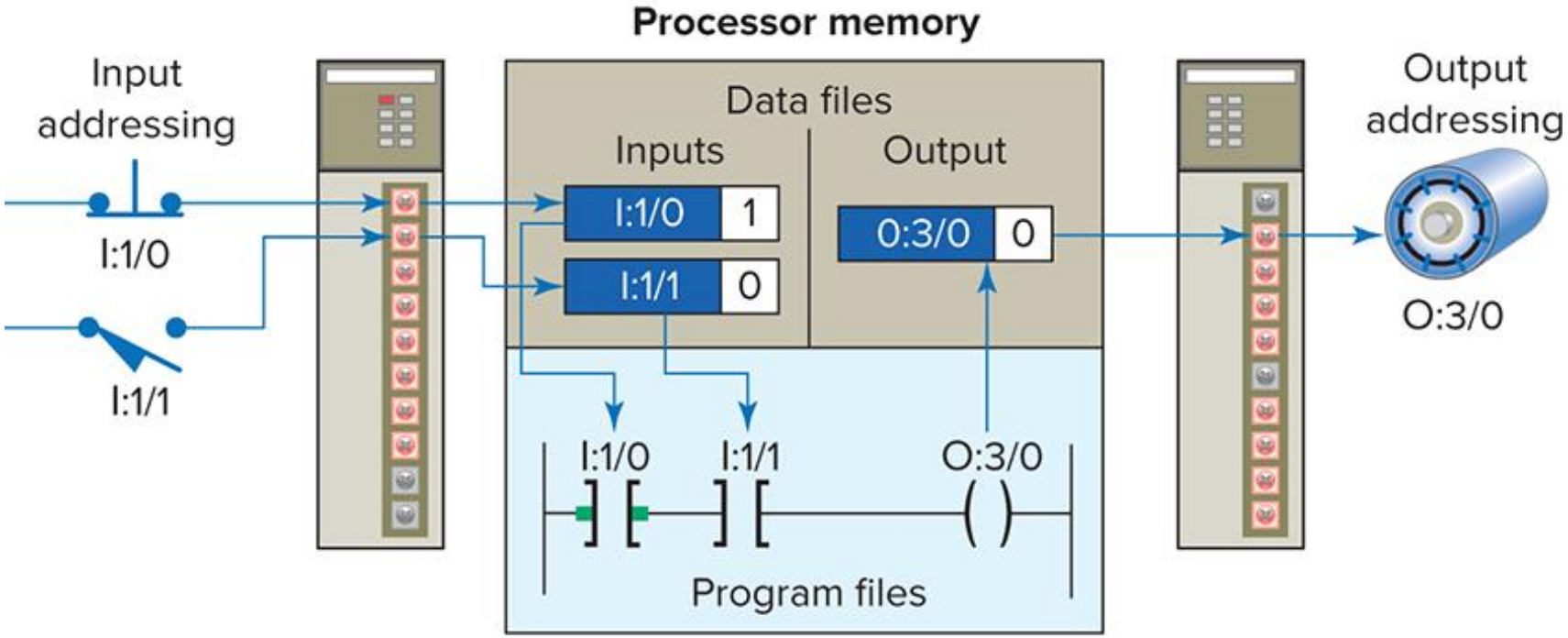
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Source: Image Courtesy of Rockwell Automation, Inc.

Bit Addressing

Every device connected to a *discrete I/O module* is addressed to a specific *bit* in the PLC's memory.



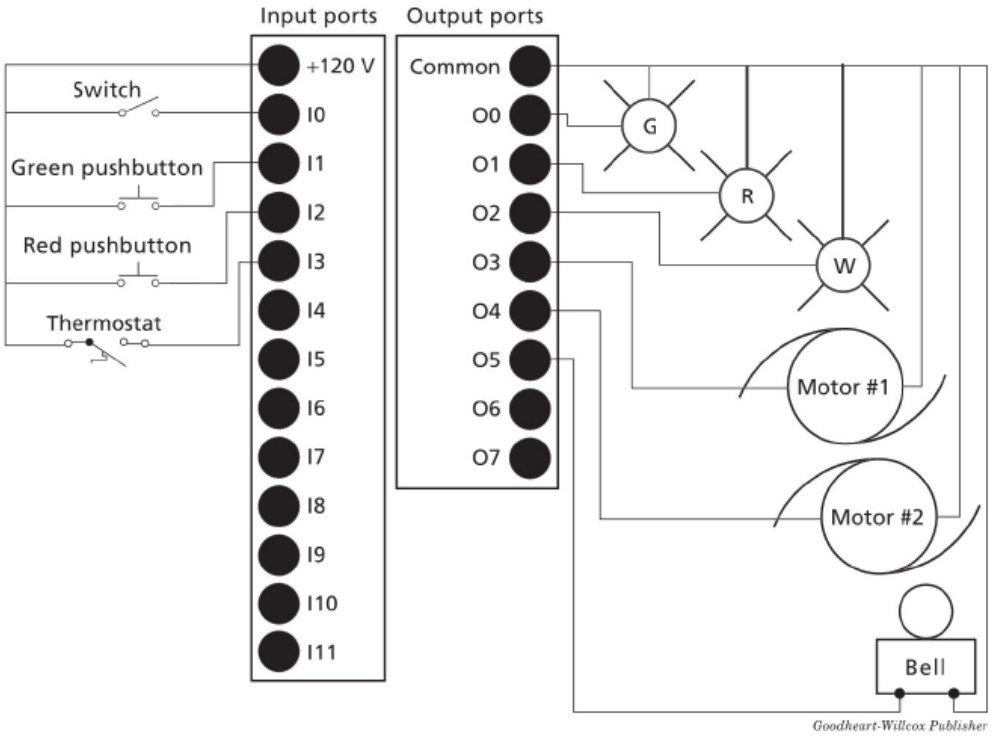
A bit is a binary digit that can be either 1 or 0.

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Addressing I/Os using A-B SLC 500

- For example, consider the following connection where a 16-port input and output modules placed in slots one and two, respectively.



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Addressing I/Os using A-B SLC 500

- For the above connection, we have the following directions

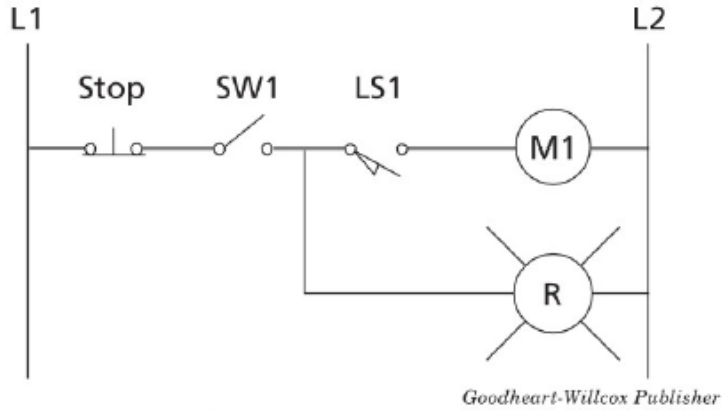
Input Addresses				Output Addresses			
I:1/0	I:1/4	I:1/8	I:1/12	O:2/0	O:2/4	O:2/8	O:2/12
I:1/1	I:1/5	I:1/9	I:1/13	O:2/1	O:2/5	O:2/9	O:2/13
I:1/2	I:1/6	I:1/10	I:1/14	O:2/2	O:2/6	O:2/10	O:2/14
I:1/3	I:1/7	I:1/11	I:1/15	O:2/3	O:2/7	O:2/11	O:2/15

- We have, then, the following addressing for the I/O devices:

Input Ports	Output Ports
Switch = I:1/0 (input port zero on module one)	Green light = O:2/0 (output port zero on module two)
Green pushbutton = I:1/1 (input port one on module one)	Red light = O:2/1 (output port one on module two)
Red pushbutton = I:1/2 (input port two on module one)	White light = O:2/2 (output port two on module two)
Thermostat = I:1/3 (input port three on module one)	Motor #1 = O:2/3 (output port three on module two)
	Motor #2 = O:2/4 (output port four on module two)
	Bell = O:2/5 (output port five on module two)

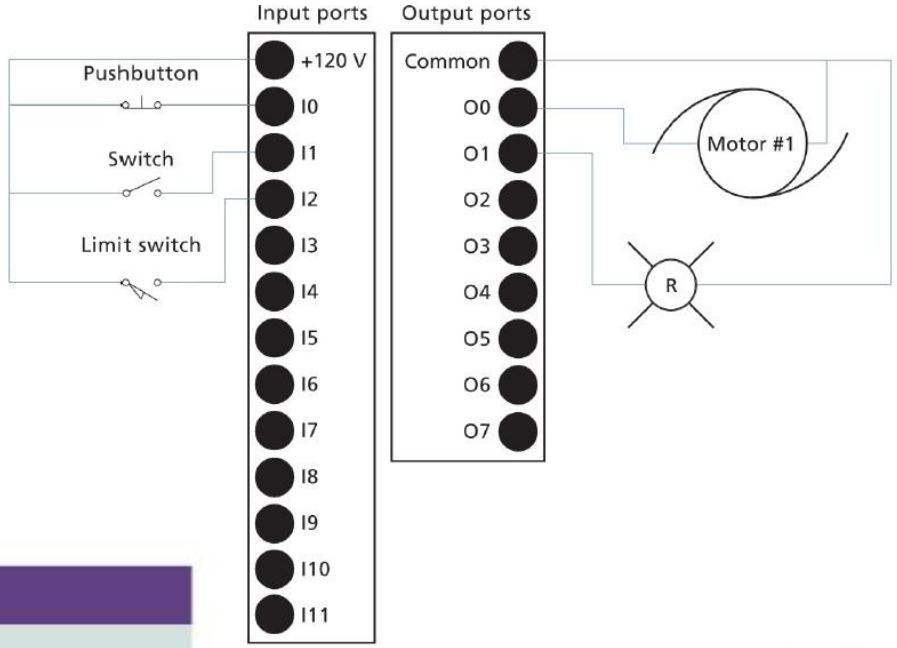
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From Relay Logic Diagram to PLC Ladder Diagram (Example 1)



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Figure 6-9. Relay logic diagram.



Input Ports	Output Ports
NC stop pushbutton = I:0/0	Motor #1 = O:0/0
NO start switch = I:0/1	Red light = O:0/1
Limit switch = I:0/2	

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From Relay Logic Diagram to PLC Ladder Diagram (Example 1) (Cont.)

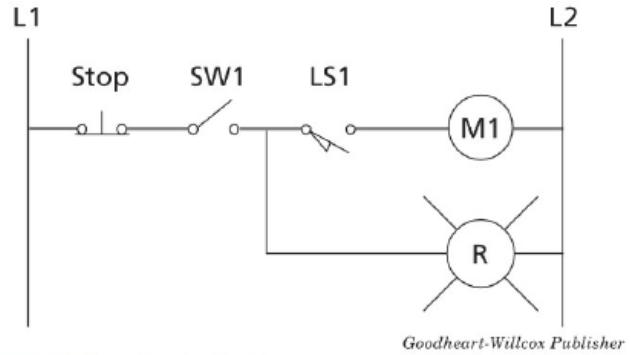
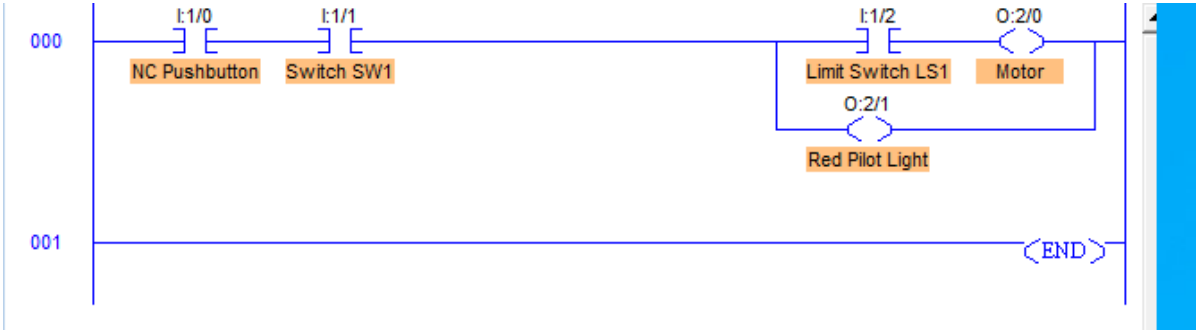


Figure 6-9. Relay logic diagram.



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Relay Diagram To PLC Diagram (Example 2)

- Relay Diagram To PLC Diagram (Example 2)

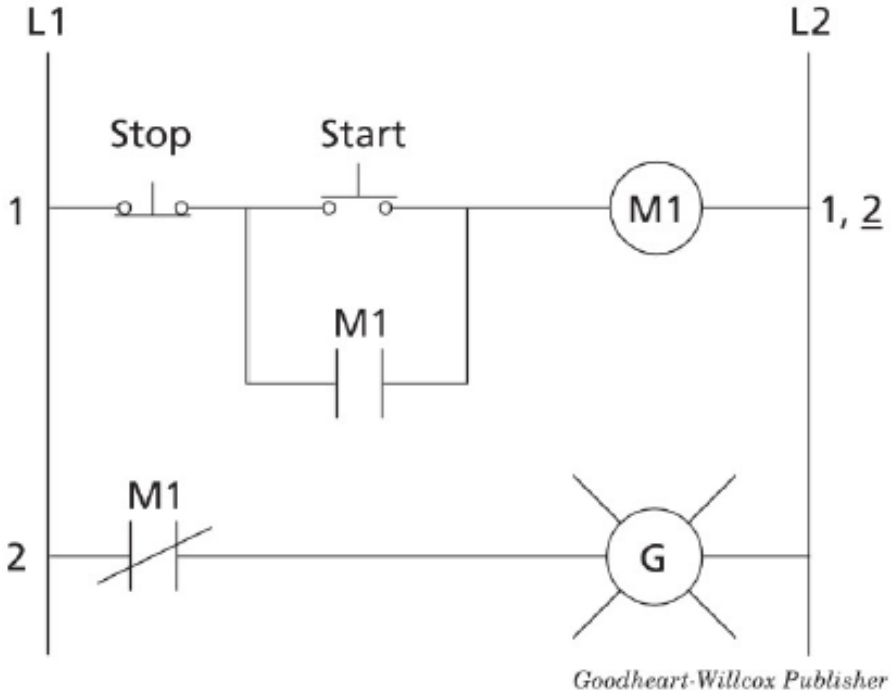
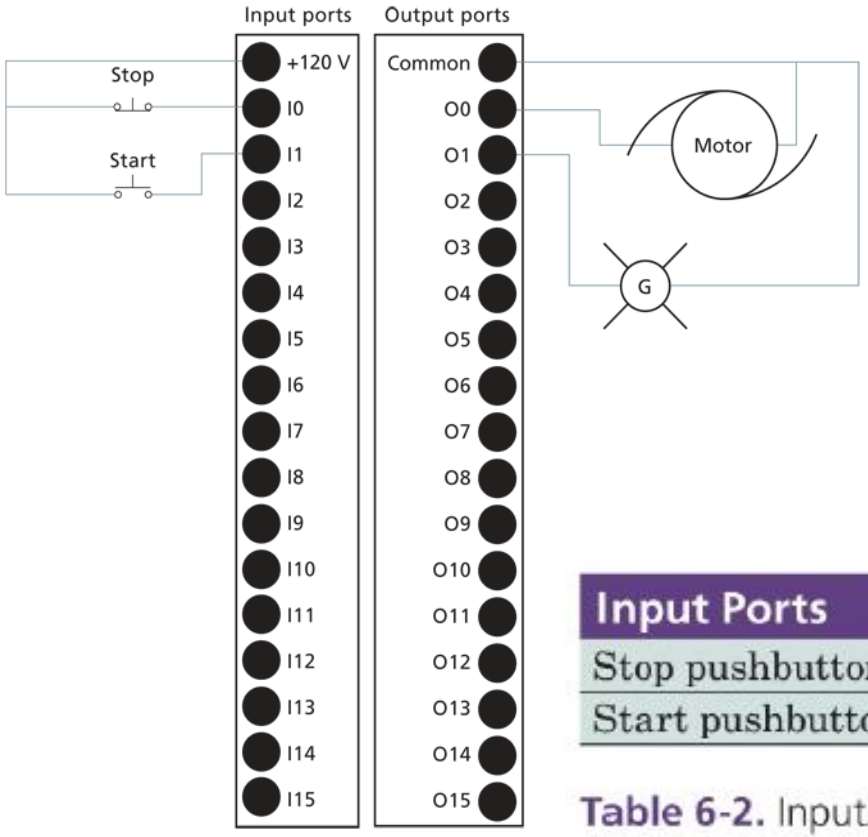
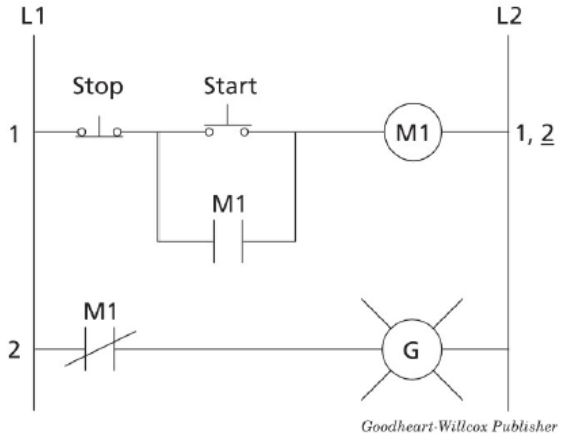


Figure 6-13. Relay logic diagram.

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Relay Diagram To PLC Diagram (Example 2)





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Figure 6-13. Relay logic diagram.

Input Ports	Output Ports
Stop pushbutton = I:1/0	Motor = O:2/0
Start pushbutton = I:1/1	Green light = O:2/1

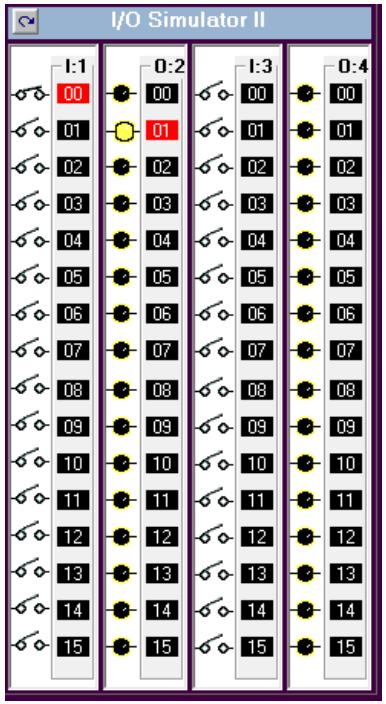
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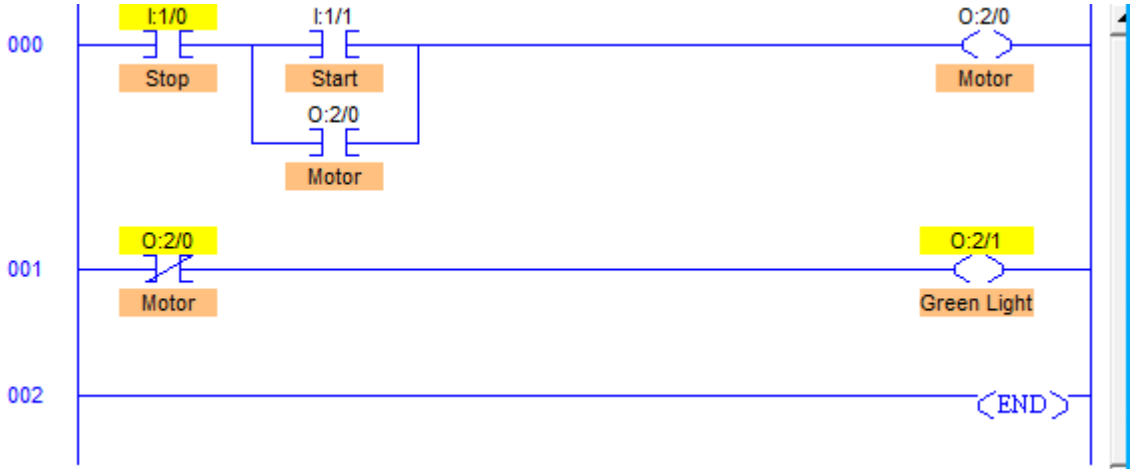
Table 6-2. Input/output address assignments for Figure 6-13.

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Relay Diagram To PLC Diagram (Example 2)

- Using the LogixPro simulator:



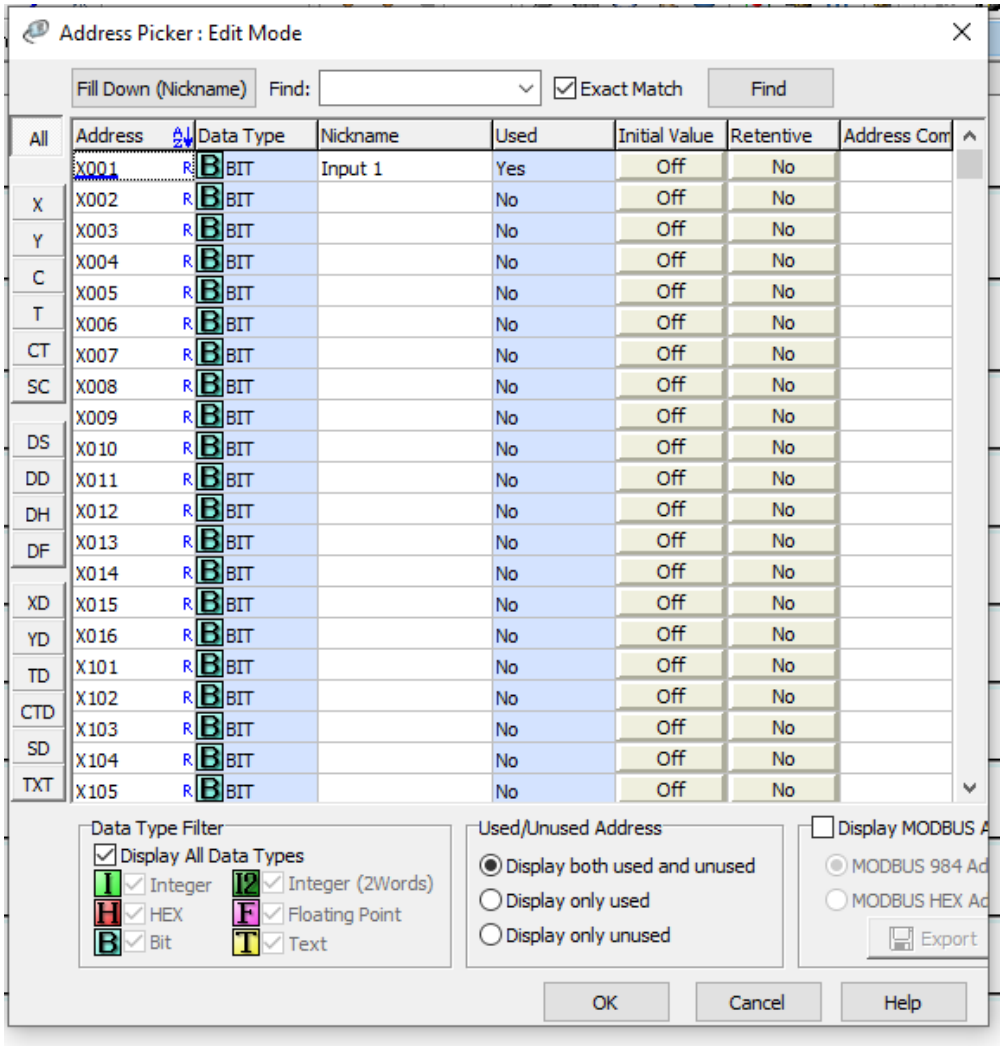


Notice that NC stop button in the relay diagram is represented with an open contact in a PLC ladder diagram, not a closed contact

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Addressing I/Os Using Click PLC

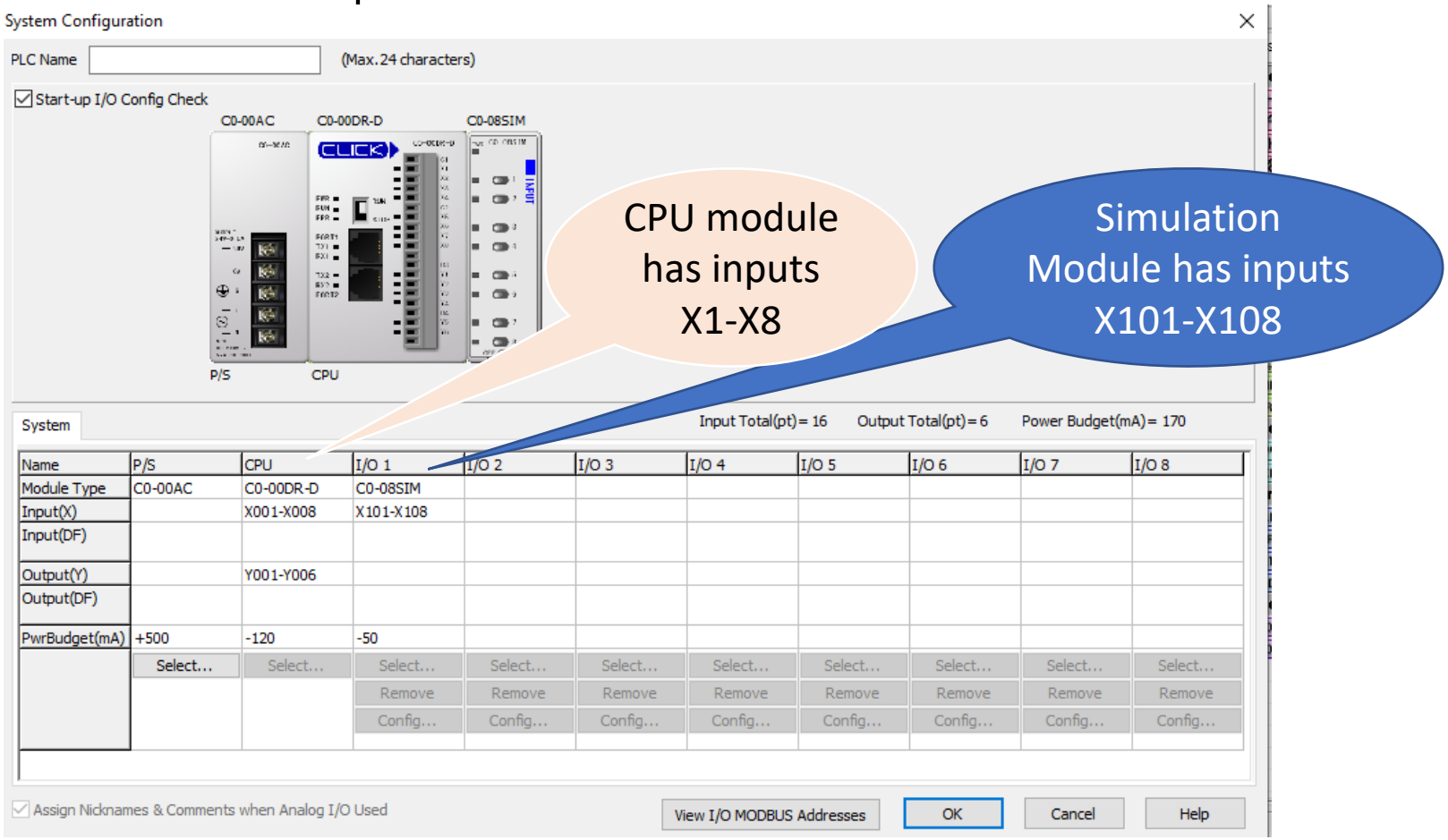
- The input addresses start with X



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Addressing I/Os using Click PLC

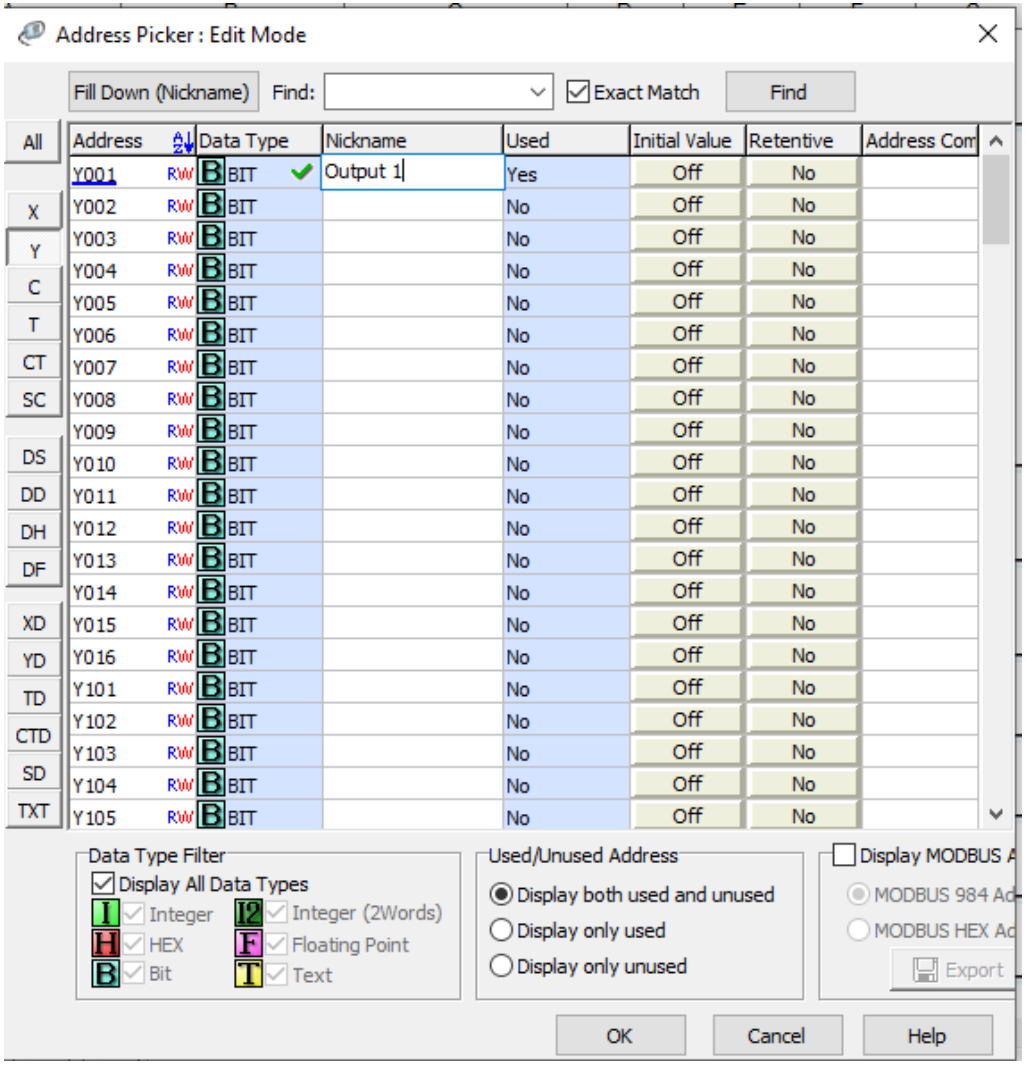
- The corresponding input number depends on the module connected to CPU. The system configuration will determine the actual number for each input.



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Addressing I/Os using Click PLC

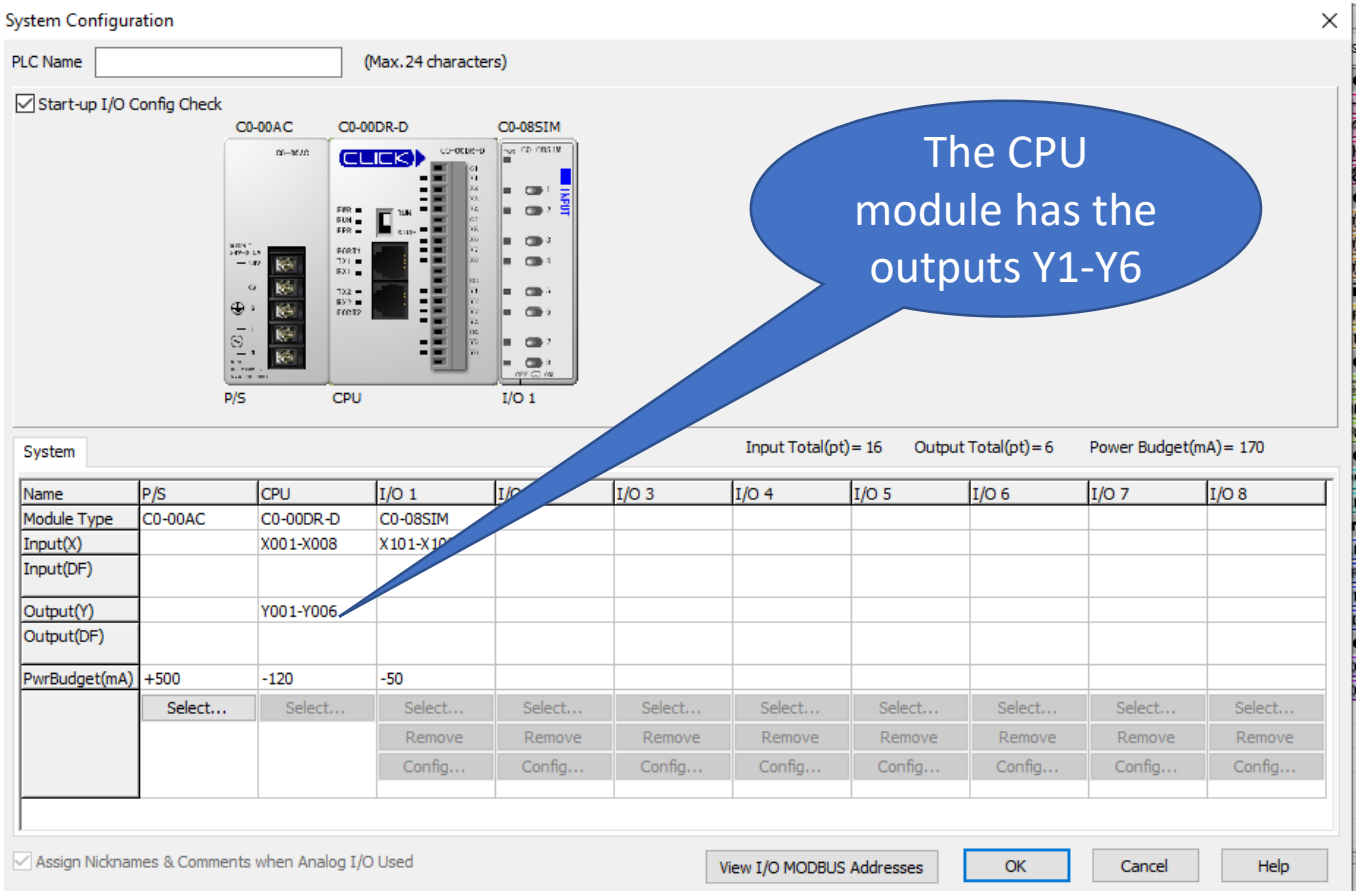
- The output addresses start with O



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Addressing I/Os using Click PLC

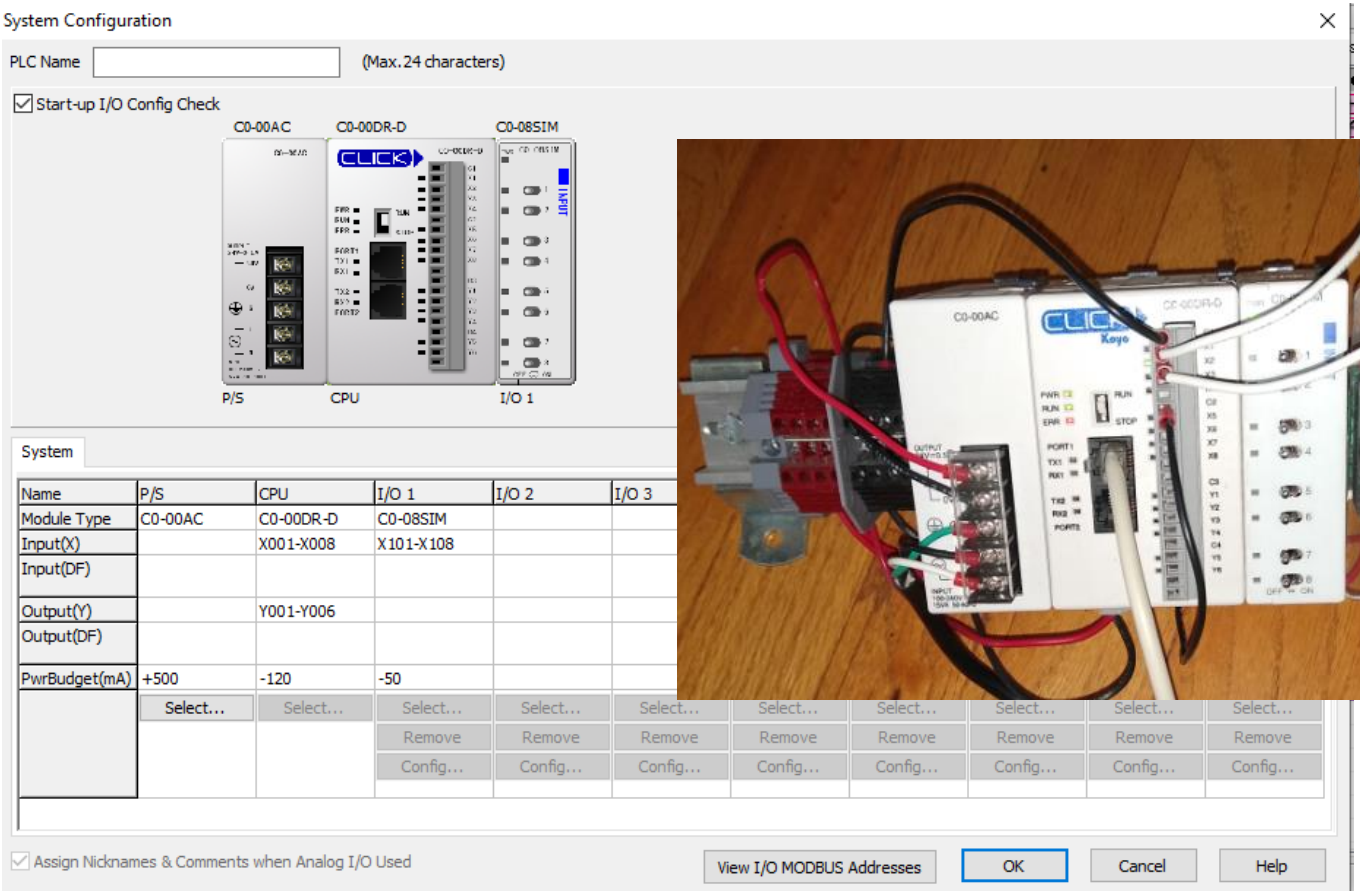
- The corresponding output number depends on the module connected to CPU. The system configuration will determine the actual number for each input.

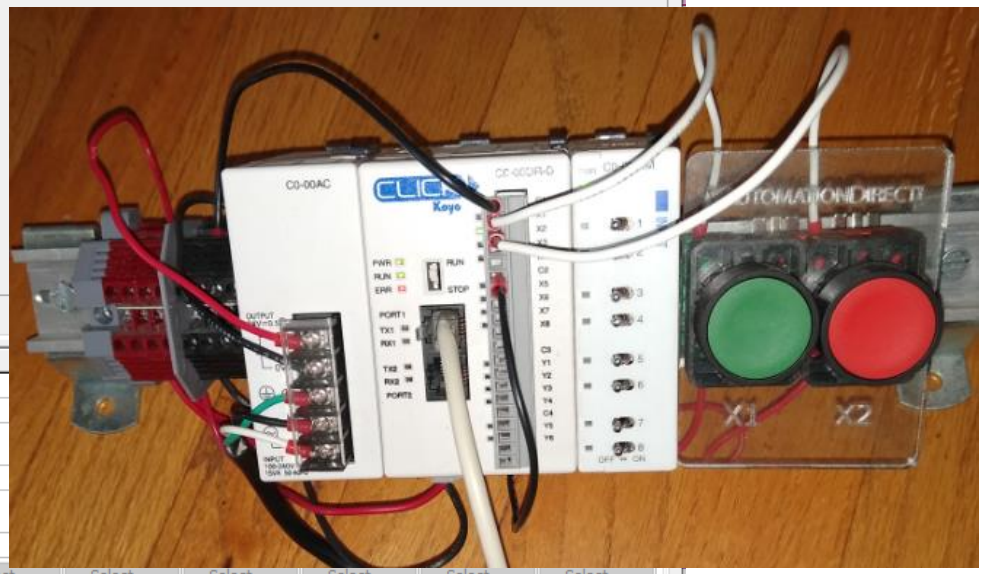


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Relay Diagram To PLC Diagram Using Click (Example 3)

- Assume we have this configuration





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Relay Diagram To PLC Diagram Using Click (Example 3)

- Notice that the NO green button (start) is connected to X1, and the NC red button (stop) is connected to X2.
- We will connect the motor to Y1, and the green indicator to Y2.
- Remember that the CPU module contains inputs X1 through X6, and outputs Y1 through Y6.
- Then the previous relay logic diagram becomes the following PLC ladder diagram in Click:

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Relay Diagram To PLC Diagram Using Click (Example 3)

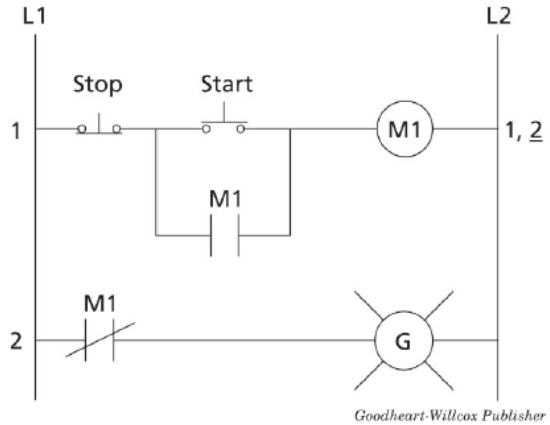
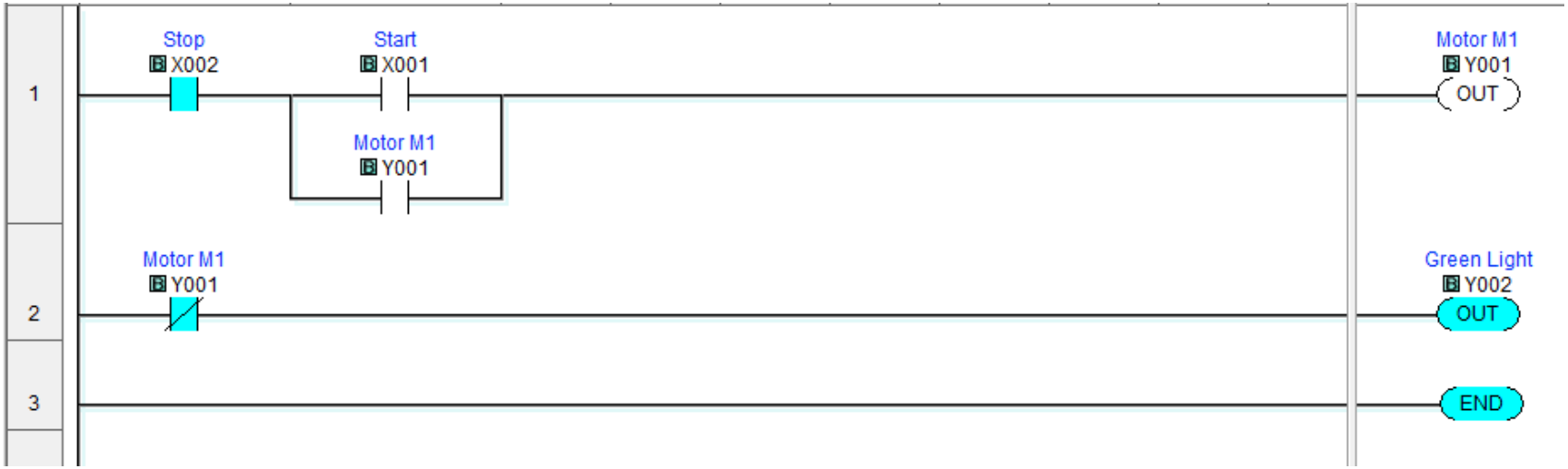


Figure 6-13. Relay logic diagram.

Notice that NC stop button in the relay diagram is represented with an open contact in a PLC ladder diagram, not a closed contact



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Rung Comments

- Placed on the top of each rung
- Describe major function of the following rungs
- Specify location of each contact associated with a coil:
  - Place contact next to the coil the number of the rung on which the associated contact is shown

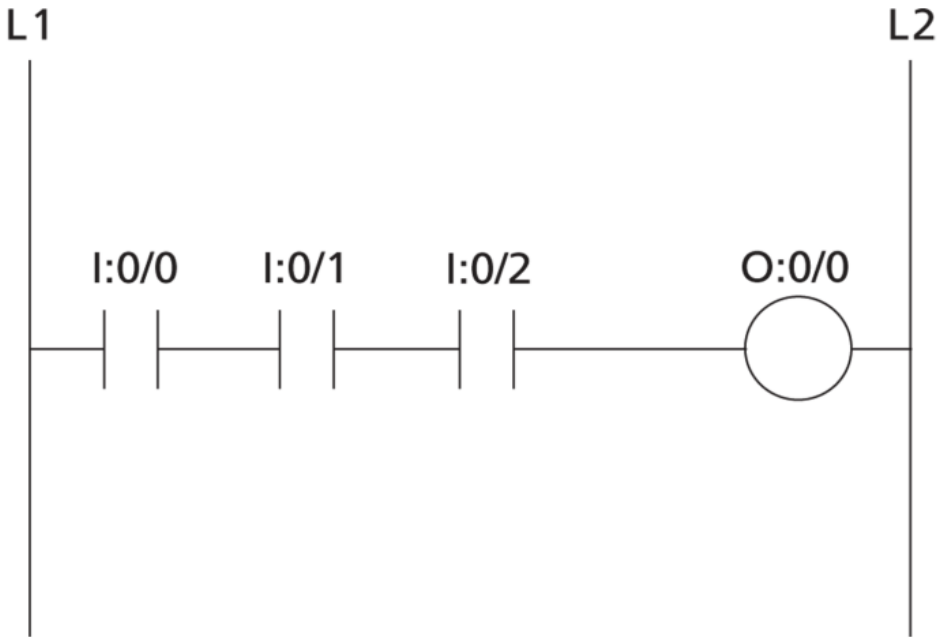
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Ladder Diagram Rules

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PLC Ladder Rules

- **Rule 1.** Do not include symbols in the PLC ladder diagram
- **Rule 2.** Place a contact in the upper-left corner of the ladder logic diagram
- **Rule 3.** Place the output (only one) at the end of the rung



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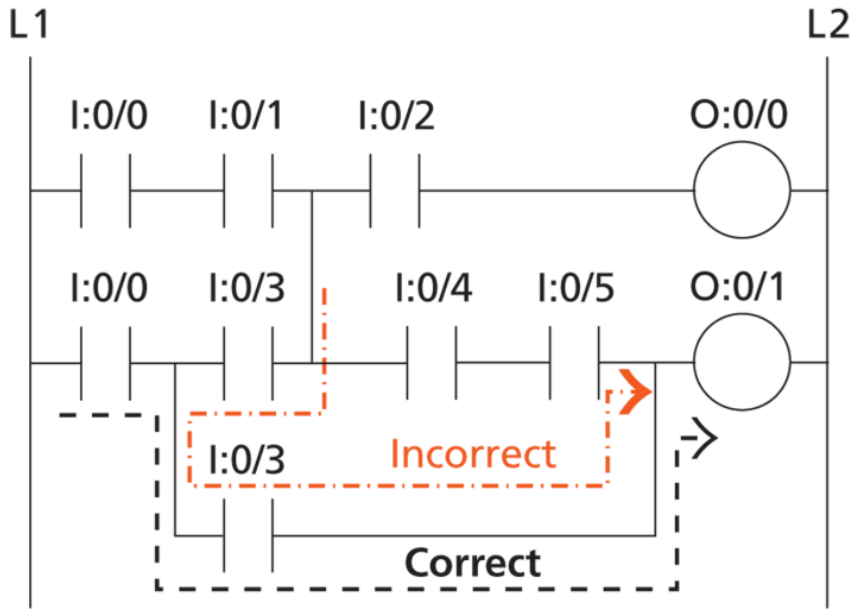
PLC Ladder Rules

- **Rule 4.** All contacts must be placed horizontally
- **Rule 5.** Outputs cannot be connected in series with other outputs
- **Rule 6.** Place the rung numbers on the left side of each rung
- **Rule 7**. Comment each rung and label all elements in your ladder diagram.

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PLC Ladder Rules

- **Rule 8.** Program execution flow must be from left to right



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